of SoC while mobile phones require another type. The type of hardware that an SoC contains depends on the usage scenario. There is no need for a graphics processor on a SoC which controls a lift while it would be an integral part of one which runs a mobile phone. We are going to list the most common ones here and explain what they do and where they can be found.

Microprocessor

Microprocessor, sometimes referred to as just the 'processor' is a small, generic processor with all the functional units. Microprocessors have their own instruction set using which they can run generic programs. They do not have a specific function (like a DSP, which is tuned for only processing signals), instead they can take a series of instructions and can execute them one after another.

Any SoC which is supposed to execute generic programs would contain them. Such SoCs can be used to execute application programs. Typically, a microprocessor needs an OS to function. It need not be a full-fledged desktop-grade OS and can be something that fits in the ROM (memory of the microprocessor). Depending on the capabilities of the entire SoC, it can be a feature-rich OS as well (think Android or iOS).

If this made you think about mobile phones, then you are right. They are also found on music players, smartwatches and miniature-size PCs (like Raspberry Pi).

Microcontroller

Microcontrollers are a superset of microprocessors from a design perspective. A Microcontroller normally consists of a microprocessor, some ROM and some RAM memory along with programmable I/O controllers. The way microcontrollers normally differ from microprocessors is:

- The microprocessor included with a microcontroller may not have a wide variety of functionality found in common gadgets like smartphones or smartwatches. They lack and extensive instruction set and features.
- The amount of RAM and ROM are very less compared to what we are used to in our day-to-day gadgets. They are just enough for the use-case they are designed for.

Microcontrollers can be generally found in appliances. Washing machines, music systems, ovens, refrigerators, car monitoring systems are fine examples of microcontroller-based SoCs. They do not run full-size OSes